



BUTACU IONEL-CATALIN

[linkedin.com/in/catalin-butacu](https://www.linkedin.com/in/catalin-butacu)

github.com/CatalinButacu

butacu.catalin@yahoo.com

+40 753 320 964

SUMMARY

C++ Software Engineer with 3+ years of professional experience developing desktop application features in Qt on a commercial industrial CAD/CAM product. Background in control systems engineering and applied informatics. Currently completing a Master's degree in Artificial Intelligence.

PROFESSIONAL EXPERIENCE

Gemini CAD Systems

Iași, RO

Software Engineer

June 2023 – Present

- Built features spanning Qt's UI, networking, model/view, and event-handling layers for real time applications.
- Designed scalable software components by applying core OOP principles and industry-standard design patterns.
- Utilized C++ STL containers and algorithms to implement efficient business logic and data processing.
- Decomposed complex requirements into scoped deliverables, consistently meeting sprint estimations.
- Reduced technical debt through targeted refactoring and clean code, improving codebase resilience.
- Diagnosed and resolved production crashes and memory leaks using debugging tools and root cause analysis.
- Participated in the full Scrum lifecycle: sprint planning, estimation, code reviews, sprint demos, and retrospectives.

Software Engineer Intern

August 2022 – May 2023

- Built UI components from design specifications, integrating them into existing production views.
- Delivered pixel-accurate UI and extended feature parameters through import/export integrations.
- Practised all Git workflow stages: branching, commits, merge requests, code reviews, and CI/CD.

Special Telecommunications Service (STS)

Botoșani, RO

IT Technician – Electoral Bureau

August 2020 – September 2020

- Supported the electoral process by ensuring IT infrastructure was operational at electoral bureaus.
- Assisted in verifying and transmitting electoral data through IT systems (SIMPV/SICPV).
- Offered technical assistance and guidance to electoral bureau staff.

CERTIFICATIONS

- Oracle Cloud Infrastructure 2025 Data Science Professional (1Z0-1110-25) [🔗](#)
- Oracle Cloud Infrastructure 2025 Generative AI Professional (1Z0-1127-25) [🔗](#)
- Oracle Cloud Infrastructure 2025 AI Vector Search Professional (1Z0-184-25) [🔗](#)
- Oracle Analytics Cloud 2025 Certified Professional (1Z0-1041-25) [🔗](#)

SKILLS & COMPETENCIES

- | | |
|--|---|
| ◦ Advanced C++ Programming (C++17/20) | ◦ Problem-Solving and Analytical Thinking |
| ◦ Qt Framework and STD Library | ◦ Technical Debt and Refactoring |
| ◦ Object-Oriented Design and Design Patterns | ◦ Memory Management, RAII, Smart Pointers |
| ◦ Multithreading and Concurrent Programming | ◦ Code Review, GitLab CI/CD, Agile/Scrum |

EDUCATION

Technical University of Iași

Iași, RO

Master of Science in Artificial Intelligence

October 2024 - June 2026

Advanced Coursework: Machine Learning, Deep Learning, Natural Language Processing (NLP), Reinforcement Learning, Multi-Agent Systems, Cloud Computing, Big Data Techniques, Edge Computing Applications, Intelligent Cyber-Security

Bachelor of Science in Computer Engineering

October 2020 - June 2024

Core Curriculum: Advanced Computer Programming, Data Structures & Algorithms, Real-Time Programming, Database Management Systems, Data Processing, Microprocessor-Based Systems, Embedded Systems Engineering, Control Engineering, Computer Networks, Numerical Methods, Neural Networks, Optimization Techniques, Fuzzy Logic, Knowledge-Based Systems

PROJECTS

- GenAI with Physics Constraints (in progress)

2026

Text-to-video pipeline that generates physics-realistic 3D animations from natural language prompts. Integrates Shap-E for 3D asset generation, motion synthesis models, and PyBullet for rigid-body physics simulation, enhanced by ControlNet.
- IDS/IPS Smart Firewall

2026

Collaborative intelligent firewall system featuring a CNN threat classifier and an RL adversarial agent. Personal contribution focused on system architecture, Snort rule configuration, and developing a DBSCAN clustering module that analyzes runtime payloads to auto-generate rules, reducing ML inference load.
- Hand Gesture Recognition

2025

YOLOv5n object-detection pipeline for hand gesture recognition (HaGRID dataset). Optimised for NPU/CPU deployment via TFLite INT8 quantisation, CLAHE contrast preprocessing, and temporal smoothing to reduce false positives.
- AI Medication Assistant

2025

Full-stack RAG application: React/TypeScript frontend (GitHub Pages) backed by a Gradio API (Hugging Face Spaces). FAISS-indexed knowledge base of 1200+ medications across 20 medical categories; answers symptom queries and surfaces direct pharmacy links.
- Metaheuristic Framework

2025

Python library implementing Canonical GA, Real-Valued GA, Differential Evolution, and Random Search. Each solver is benchmarked on standard mathematical test functions with automated CSV reporting and convergence visualisation.
- fMRI Brain-Age Prediction

2025

Competition ML pipeline predicting biological brain age from ~20k fMRI neuroimaging connectivity features. Gender-stratified ElasticNet model with spectral feature selection; ranked 8th nationally at the WiDS Datathon 2025 University Edition.

VOLUNTEER EXPERIENCE

- LOCAL BEST Group Iași

Iași, RO

IT Department Coordinator

Sep 2023 – Aug 2024

- Directed IT operations and maintained technical infrastructure across organizational events.
 - Mentored volunteers and oversaw delivery of technical projects.
 - Administered hardware, software licensing, and network resources.
- Event Coordinator | JobShop Career Event

Nov 2022 – Apr 2023

- Coordinated logistics and operations for a large student–industry career event.
 - Managed scheduling, partner engagement, and on-site execution.
 - Improved attendee experience through strategic planning and workflow optimization.
- TOPICS Responsible | Prove It 24H Techathon

Dec 2021 – Mar 2022

- Defined technical requirements and coordinated hackathon logistics.
 - Supported participants with rules, resources, and real-time assistance.
 - Facilitated communication between organizers, industry mentors, and teams.
- IT Responsible | BEST Training Weekend

Jan 2021 – May 2022

- Developed and maintained the event website using HTML, CSS, and JavaScript.
 - Collaborated with design and marketing responsables on content, branding, and technical setup.

ADDITIONAL SKILLS

- AI / ML:

Machine Learning, Deep Learning, LLMs, RAG Systems, PyTorch/TF, Prompt Engineering
- Web Apps:

Flask, Hadoop, React, TypeScript, Node.js, Firebase, PostgreSQL, WordPress
- Soft Skills:

Problem Decomposition, Clear Communication, Knowledge Sharing, Self-directed Learning